Design and Evaluation Report for the Electronics II Project (Phase II)

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1 Project Requirements

Main Phase-II Requirements

In this phase, the completed amplifier must be designed and evaluated as a single integrated circuit containing a differential input stage, voltage-amplification stages, bias networks, a power output stage, the output load, and global negative feedback. The final circuit must satisfy the following requirements:

Required Value	Parameter
50Ω	Output load resistance (R_L)
$18 \leq \lvert A_{v,\text{closed}} \rvert \leq 22$	Closed-loop gain at 1 kHz
$10k\Omega$ to $200k\Omega$	Allowed range of R_f and R_g
$16V_{pp}$	Main output swing without severe clipping
$> 60\%$	Output-stage efficiency at $16V_{pp}$
$< 0.08\%$	THD under normal conditions
$< 1\%$	THD in the presence of the specified noise sources
$\leq 190mW$	Total power at $50mV$ and 1 kHz input
$> 1M\Omega$	Differential input resistance ($R_{in,\text{diff}}$)
$< 50\Omega$	Output resistance (R_{out}), with R_L removed
$10mV$	Sawtooth supply ripple in the PSRR test
≤ 200	Maximum circuit cost

Final Specification Check

The final measured results are summarized below. All main tests were performed on the complete feedback amplifier with $R_L = 50\Omega$, except for the output-resistance test, in which the load was removed as required.

Parameter	Requirement	Measured Value	Status
$\lvert A_{v,\text{closed}}\rvert$	18 to 22	20.877	**PASS**
$V_{out,pp}$	at least $16V_{pp}$	$16.050V_{pp}$	**PASS**
η_{out}	$> 60\%$	63.33%	**PASS**
P_{total} at $50mV$ input	$\leq 190mW$	$163.759mW$	**PASS**
Normal THD	$< 0.08\%$	0.040374%	**PASS**
THD with noise	$< 1\%$	0.819073%	**PASS**
PSRR	report the value	$45.429dB$	–
$R_{in,\text{diff}}$	$> 1M\Omega$	$32.479M\Omega$	**PASS**
R_{out}	$< 50\Omega$	0.2135Ω	**PASS**
Cost	≤ 200	190	**PASS**

2 Final Circuit and Overall Architecture

Required Work (Part 1)

Introduce the final project circuit and state that only one complete final amplifier is submitted. The circuit must include the differential input, voltage-amplification stages, bias circuits, power output stage, $R_L = 50\Omega$ load, and global negative-feedback network. Also explain the order of completion: first adding the output stage and then closing the global feedback loop.

Selected Final Architecture

The final design is a multistage amplifier powered from symmetric $\pm 10V$ supplies and implemented as one integrated circuit. Its overall structure consists of the input differential pair Q_1–Q_2, active loads and current mirrors, voltage-amplification stages Q_6 through Q_{10}, bias networks, a complementary class-AB output stage, and a resistive global-feedback network.

The output stage is formed by the complementary emitter followers Q_{13} and Q_{16} and is driven and biased by Q_{10}, Q_{11}, Q_{12}, Q_{17}, and Q_{18}. The bias voltage between the two output-transistor bases is set close to two base-emitter drops, thereby reducing crossover distortion. The 50Ω load is connected directly to the final output node.

The output voltage is sampled by $R_f = 200k\Omega$ and returned to the base of Q_1. Resistor $R_g = 10k\Omega$ connects the same node to ground, while the external input is applied to the base of Q_2; therefore, the overall configuration is non-inverting. The power output stage was connected and biased first, and the global feedback loop was closed only after the output stage had been adjusted.

A compensation capacitor $C_1 = 82pF$ was placed around the Q_{10} voltage-amplification stage to improve loop stability and suppress undesired high-frequency components. Compared with Phase I, the principal additions are the power stage, the class-AB bias network, the 50Ω load, global negative feedback, and Miller compensation.

Complete Final Schematic

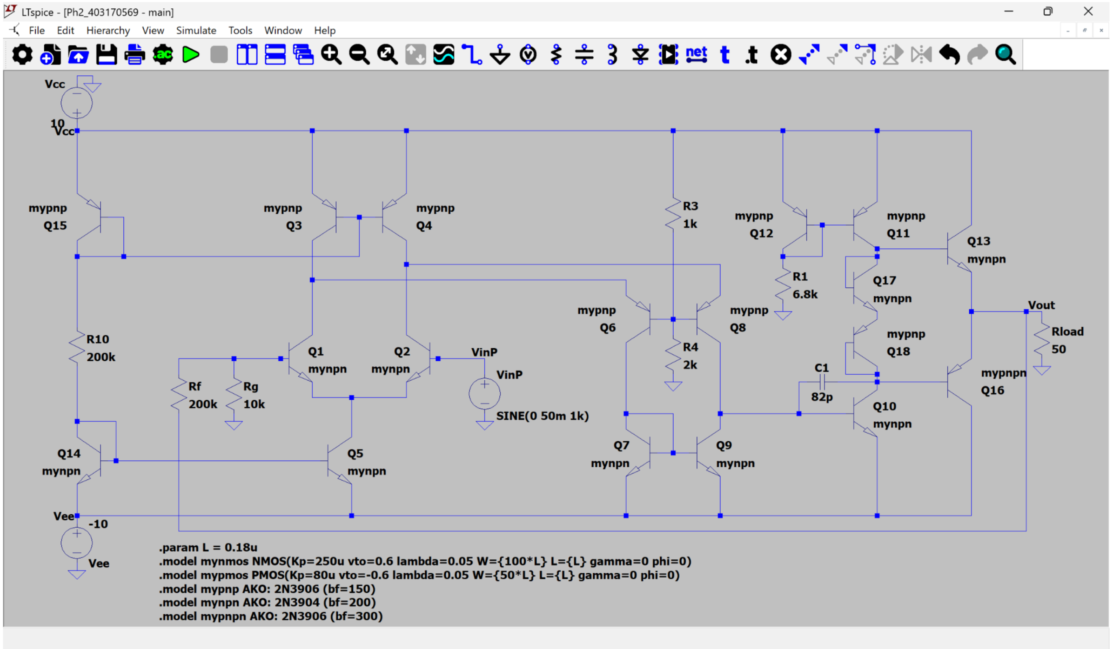


Figure 1 – Complete final schematic including the power stage, 50Ω load, global feedback, and the $82pF$ compensation capacitor.

The principal final values are $R_f = 200k\Omega$, $R_g = 10k\Omega$, $R_1 = 6.8k\Omega$, $R_3 = 1k\Omega$, $R_4 = 2k\Omega$, $R_{10} = 200k\Omega$, and $C_1 = 82pF$.

3 DC Analysis and Biasing of the Final Circuit

Required Work (Part 2)

Perform a DC analysis of the final circuit. Evaluate the operating points of the principal stages, the bias currents, important node voltages, and the output-stage bias condition. If a V_{BE} multiplier or an equivalent network is used, report its bias voltage.

Theoretical Bias Calculations

The operating-point analysis was performed with zero input and a 50Ω load. From the DC results, the reference current in the left branch is

$$I_{R10} = 94.06\,\mu A.$$

The tail current of the differential pair, supplied by Q_5, is approximately $101.35\,\mu A$ and is divided as

$$I_{C,Q1} = 53.64\,\mu A, \qquad I_{C,Q2} = 47.24\,\mu A.$$

The small mismatch is caused by input bias currents and the closed-loop offset.
For the middle bias network,

$$I_{R3} = \frac{10 - 6.667}{1k} = 3.333\,mA, \qquad I_{R4} = \frac{6.667}{2k} = 3.333\,mA.$$

The output-stage bias branch current through R_1 is approximately

$$I_{R1} = 1.373\,mA.$$

At the operating point, the two output-drive node voltages are

$$V_{BUP} = 0.8025V, \qquad V_{BDN} = -0.5277V,$$

and hence the bias voltage across the output-transistor bases is

$$V_{bias} = V_{BUP} - V_{BDN} = 1.3302V.$$

This is close to two base-emitter junction drops and therefore establishes class-AB operation.
The output offset is $V_{out,DC} = 119.087mV$. With the 50Ω load, the DC load current is

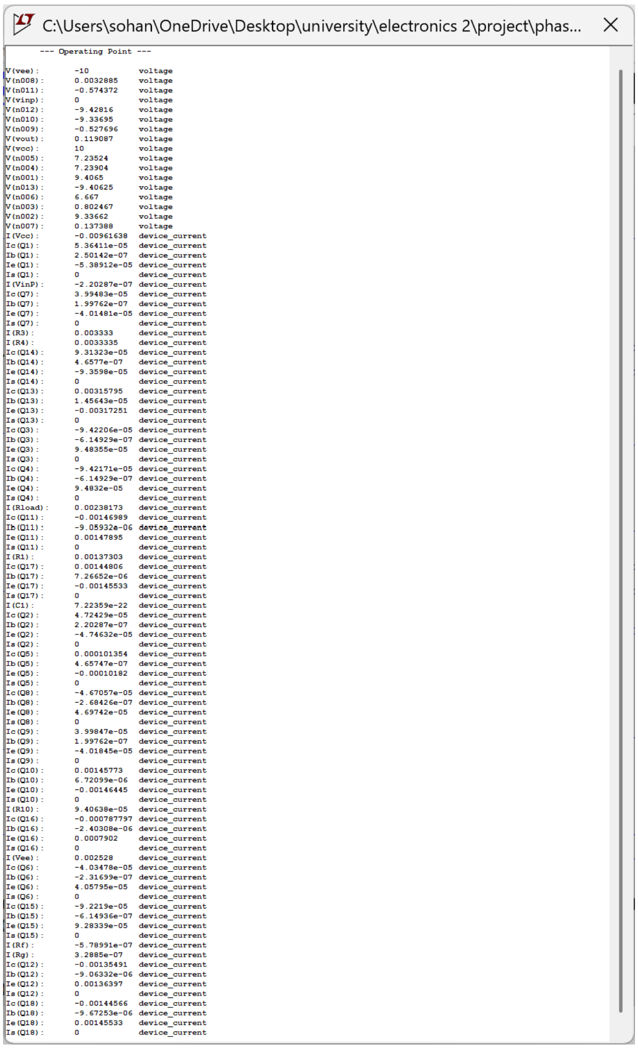
$$I_{RL,DC} = \frac{0.119087}{50} = 2.382mA.$$

The output-transistor collector currents at the operating point are

$$I_{C,Q13} = 3.158mA, \qquad |I_{C,Q16}| = 0.779mA.$$

Their difference supplies the DC current delivered to the load.

DC Operating-Point Simulation Results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--- Operating Point ---

V(vee):	-10	voltage
V(n008):	0.0032885	voltage
V(n011):	-0.574372	voltage
V(vinp):	0	voltage
V(n012):	-9.42816	voltage
V(n010):	-9.33695	voltage
V(n009):	-0.527696	voltage
V(vout):	0.119087	voltage
V(vcc):	10	voltage
V(n005):	7.23524	voltage
V(n004):	7.23904	voltage
V(n001):	9.4065	voltage
V(n013):	-9.40625	voltage
V(n006):	6.667	voltage
V(n003):	0.802467	voltage
V(n002):	9.33662	voltage
V(n007):	0.137388	voltage
I(Vcc):	-0.00961638	device_current
Ic(Q1):	5.36411e-05	device_current
Ib(Q1):	2.50142e-07	device_current
Ie(Q1):	-5.38912e-05	device_current
Is(Q1):	0	device_current
I(VinP):	-2.20287e-07	device_current
Ic(Q7):	3.99483e-05	device_current
Ib(Q7):	1.99762e-07	device_current
Ie(Q7):	-4.01481e-05	device_current
Is(Q7):	0	device_current
I(R3):	0.003333	device_current
I(R4):	0.0033335	device_current
Ic(Q14):	9.31323e-05	device_current
Ib(Q14):	4.6577e-07	device_current
Ie(Q14):	-9.3598e-05	device_current
Is(Q14):	0	device_current
Ic(Q13):	0.00315795	device_current
Ib(Q13):	1.45643e-05	device_current
Ie(Q13):	-0.00317251	device_current
Is(Q13):	0	device_current
Ic(Q3):	-9.42206e-05	device_current
Ib(Q3):	-6.14929e-07	device_current
Ie(Q3):	9.48355e-05	device_current
Is(Q3):	0	device_current
Ic(Q4):	-9.42171e-05	device_current
Ib(Q4):	-6.14929e-07	device_current
Ie(Q4):	9.4832e-05	device_current
Is(Q4):	0	device_current
I(Rload):	0.00238173	device_current
Ic(Q11):	-0.00146989	device_current
Ib(Q11):	-9.05932e-06	device_current
Ie(Q11):	0.00147895	device_current
Is(Q11):	0	device_current
I(R1):	0.00137303	device_current
Ic(Q17):	0.00144806	device_current
Ib(Q17):	7.26652e-06	device_current
Ie(Q17):	-0.00145533	device_current
Is(Q17):	0	device_current
I(C1):	7.22359e-22	device_current
Ic(Q2):	4.72429e-05	device_current
Ib(Q2):	2.20287e-07	device_current
Ie(Q2):	-4.74632e-05	device_current
Is(Q2):	0	device_current
Ic(Q5):	0.000101354	device_current
Ib(Q5):	4.65747e-07	device_current
Ie(Q5):	-0.00010182	device_current
Is(Q5):	0	device_current
Ic(Q8):	-4.67057e-05	device_current
Ib(Q8):	-2.68426e-07	device_current
Ie(Q8):	4.69742e-05	device_current
Is(Q8):	0	device_current
Ic(Q9):	3.99847e-05	device_current
Ib(Q9):	1.99762e-07	device_current
Ie(Q9):	-4.01845e-05	device_current
Is(Q9):	0	device_current
Ic(Q10):	0.00145773	device_current
Ib(Q10):	6.72099e-06	device_current
Ie(Q10):	-0.00146445	device_current
Is(Q10):	0	device_current
I(R10):	9.40638e-05	device_current
Ic(Q16):	-0.000787797	device_current
Ib(Q16):	-2.40308e-06	device_current
Ie(Q16):	0.0007902	device_current
Is(Q16):	0	device_current
I(Vee):	0.002528	device_current
Ic(Q6):	-4.03478e-05	device_current
Ib(Q6):	-2.31699e-07	device_current
Ie(Q6):	4.05795e-05	device_current
Is(Q6):	0	device_current
Ic(Q15):	-9.2219e-05	device_current
Ib(Q15):	-6.14936e-07	device_current
Ie(Q15):	9.28339e-05	device_current
Is(Q15):	0	device_current
I(Rf):	-5.78991e-07	device_current
I(Rg):	3.2885e-07	device_current
Ic(Q12):	-0.00135491	device_current
Ib(Q12):	-9.06332e-06	device_current
Ie(Q12):	0.00136397	device_current
Is(Q12):	0	device_current
Ic(Q18):	-0.00144566	device_current
Ib(Q18):	-9.67253e-06	device_current
Ie(Q18):	0.00145533	device_current
Is(Q18):	0	device_current

Figure 2 – LTspice DC operating-point results for the final circuit with zero input.

Important results:

- Output DC voltage at zero input: $V_{out,DC} = 0.119087V$.
- Quiescent output-transistor currents: $I_{C,Q13} = 3.158mA$ and $|I_{C,Q16}| = 0.779mA$.
- DC load current: $2.382mA$.
- Output-stage bias voltage: $1.330V$.
- Positive-supply current at the operating point: approximately $9.616mA$; negative-rail branch current: approximately $2.528mA$.

All gain-stage transistors operate in their active regions. The two-junction output bias creates a small overlap in conduction around the zero crossing and reduces crossover distortion. The approximately $119mV$ output offset does not violate any mandatory specification.

4 Power Output-Stage Design

Required Work (Part 3)

Introduce and analyze the power output stage. Show that it is connected integrally to the main amplifier and is designed to drive the 50Ω load. Explain its topology, transistor types, biasing, and the reasons for selecting this structure.

Theoretical Output-Stage Analysis

To generate a $16V_{pp}$ output across 50Ω, the required voltage and current peaks are

$$V_p = \frac{16}{2} = 8V, \qquad I_{L,p} = \frac{V_p}{R_L} = \frac{8}{50} = 160mA.$$

The corresponding RMS voltage and theoretical load power are

$$V_{L,rms} = \frac{8}{\sqrt{2}} = 5.657V, \qquad P_L = \frac{V_{L,rms}^2}{50} = 0.64W.$$

A complementary emitter-follower stage was selected because it provides high current gain, low output resistance, and a voltage swing close to the supply rails. The theoretical maximum efficiency of an ideal class-B sinusoidal stage is

$$\eta_{B,\max} = \frac{\pi}{4} \times 100\% = 78.5\%.$$

In practice, class-AB quiescent current, base-emitter drops, and driver losses reduce the attainable efficiency.
Assuming a current gain of approximately 200, the worst-case output-transistor peak base current is

$$I_{B,p} \simeq \frac{160mA}{200} = 0.8mA,$$

which can be supplied by the driver network with its approximately $1.4mA$ bias current.

Output-Stage Structure and Evaluation

The output stage is a complementary class-AB push-pull emitter follower. Q_{13} is the upper NPN device and Q_{16} is the lower PNP device; their emitters are joined at V_{out}. The stage has approximately unity voltage gain but provides the current required by the low-resistance load.
The voltage-amplification-stage output is conveyed through Q_{10} and the Q_{11}–Q_{12} current mirror to the Q_{17}–Q_{18} bias network. This network establishes approximately $1.33V$ between the bases of Q_{13} and Q_{16}, thereby reducing the crossover dead zone. The output PNP device uses a $2N3906$-based model with $\beta_F = 300$, which is within the permitted use of one different transistor model.
Capacitor $C_1 = 82pF$ is connected between the input and output of the inverting Q_{10} voltage-amplification stage and acts as a Miller compensation capacitor. It lowers loop gain before non-dominant poles introduce excessive phase lag, thereby improving phase

margin, transient damping, and overall stability. The driver network was also designed to preserve sufficient voltage headroom for an approximately $\pm 8V$ output and sufficient base current for the $160mA$ load-current peak.

Mathematical Design and Selection of the Miller Compensation Capacitor

The capacitor C_1 is connected between the base and collector of Q_{10}. Since this is an inverting gain stage, let its small-signal voltage gain be

$$A_2 = \frac{v_{c10}}{v_{b10}} < 0.$$

The current drawn by the capacitor from the input side is

$$i_C = C_1 \frac{d(v_{b10} - v_{c10})}{dt} = C_1(1 - A_2)\frac{dv_{b10}}{dt}.$$

Therefore, by Miller's theorem, the effective capacitance seen at the input of the voltage-amplification stage is approximately

$$C_{M,in} = C_1(1 - A_2) \simeq C_1(1 + |A_2|),$$

whereas the effective capacitance at the output side is approximately

$$C_{M,out} = C_1\left(1 - \frac{1}{A_2}\right) \simeq C_1.$$

Consequently, the dominant pole associated with the input of Q_{10} moves to a lower frequency:

$$f_{p1} \simeq \frac{1}{2\pi R_{in,VAS}\left[C_{\pi,Q10} + C_1(1 + |A_2|)\right]}.$$

This produces pole splitting: around the loop crossover frequency, the loop gain is dominated by one pole and falls at approximately $-20dB/dec$. Thus, the loop gain reaches unity before the additional phase lag from non-dominant poles can cause ringing or oscillation.

For a first-order estimate of C_1, the standard two-stage compensated-amplifier relation is used:

$$f_u \simeq \frac{g_{m,in}}{2\pi C_1},$$

where f_u is the loop unity-gain frequency and $g_{m,in}$ is the effective transconductance of the input stage. From the operating point,

$$I_{C,Q1} = 53.64\,\mu A, \qquad V_T \simeq 25.9mV,$$

so

$$g_{m,in} \simeq \frac{I_{C,Q1}}{V_T} = \frac{53.64\,\mu A}{25.9mV} = 2.07mS.$$

The measured closed-loop gain is $A_{CL} = 20.877$. To keep the closed-loop bandwidth at roughly ten times the highest audio frequency of $20kHz$, the target was selected as

$$f_{BW,CL} \simeq 200kHz.$$

Using $f_{BW,CL} \simeq f_u/A_{CL}$ gives

$$C_{1,calc} \simeq \frac{g_{m,in}}{2\pi A_{CL} f_{BW,CL}} = \frac{2.07mS}{2\pi(20.877)(200kHz)} = 78.9pF.$$

The nearest standard and readily available value is therefore

$$\boxed{C_1 = 82pF}.$$

With this value,

$$f_u \simeq \frac{2.07mS}{2\pi(82pF)} = 4.02MHz,$$

$$f_{BW,CL} \simeq \frac{4.02MHz}{20.877} = 192.5kHz.$$

Thus, the estimated closed-loop bandwidth is approximately 9.6 times $20kHz$, which comfortably covers the audio band with little attenuation.
The compensation capacitor also limits slew rate. Using the differential-pair tail current,

$$I_{tail} \simeq 101.35\,\mu A,$$

the first-order available slew rate is

$$SR_{available} \simeq \frac{I_{tail}}{C_1} = \frac{101.35\,\mu A}{82pF} = 1.24\,V/\mu s.$$

For a full-amplitude sinusoid with $8V$ peak at $20kHz$, the minimum required slew rate is

$$SR_{required} = 2\pi f_{max} V_p = 2\pi(20kHz)(8V) = 1.01\,V/\mu s.$$

Therefore, $82pF$ still provides sufficient slew-rate margin for a full-amplitude $20kHz$ signal. Equivalently, the approximate maximum acceptable capacitor from the slew-rate condition is

$$C_{1,max} \simeq \frac{I_{tail}}{SR_{required}} = 100.8pF.$$

Hence, $82pF$ lies in a suitable range according to both bandwidth and slew-rate constraints.
The effect of changing the capacitor can be summarized as follows:

- A smaller C_1 increases bandwidth and slew rate, but moves the loop crossover closer to non-dominant poles, reduces phase margin, and increases the probability of ringing, oscillation, and elevated noise-related THD.
- An excessively large C_1 improves stability and attenuates high-frequency components, but reduces bandwidth, transient speed, and the ability to reproduce full-amplitude high-frequency audio signals.
- The selected $82pF$ value is a compromise between these effects. It yields stable transient behavior, $THD = 0.040374\%$ without noise, and $THD = 0.819073\%$ with the specified noise sources, while maintaining the required gain, swing, and efficiency.

5 Application of Global Negative Feedback

Required Work (Part 4)

After adding the output stage, apply global negative feedback around the complete amplifier. The feedback signal must be sampled from the final output and returned to the differential input. The principal feedback network must contain only R_f and R_g, and the closed-loop gain should be approximately 20.

Feedback-Network Calculations

The closed-loop configuration is non-inverting. Assuming sufficiently large open-loop gain,

$$A_{v,CL} \simeq 1 + \frac{R_f}{R_g}.$$

With

$$R_f = 200k\Omega, \qquad R_g = 10k\Omega,$$

we obtain

$$A_{v,CL,ideal} = 1 + \frac{200k}{10k} = 21.$$

Both resistors lie within the allowed $10k\Omega$ to $200k\Omega$ range. The feedback factor is

$$\beta = \frac{R_g}{R_f + R_g} = \frac{1}{21} = 0.04762.$$

In the standard simulation, $0.1V_{pp}$ was applied and $2.08775V_{pp}$ was measured at the output, so

$$|A_{v,CL}| = \frac{2.08775}{0.1} = 20.8775,$$

or

$$20\log_{10}|A_{v,CL}| = 26.39dB.$$

The error relative to the ideal value is only about 0.58%.

Feedback Results

Final values and connections:

- $R_f = 200k\Omega$.
- $R_g = 10k\Omega$.
- Sampling point: final V_{out} node, after the power stage and in the presence of the load.
- Feedback injection point: base of Q_1; the base of Q_2 is the signal input.
- Closed-loop gain at $1kHz$: $|A_v| = 20.8775$ or $26.39dB$.

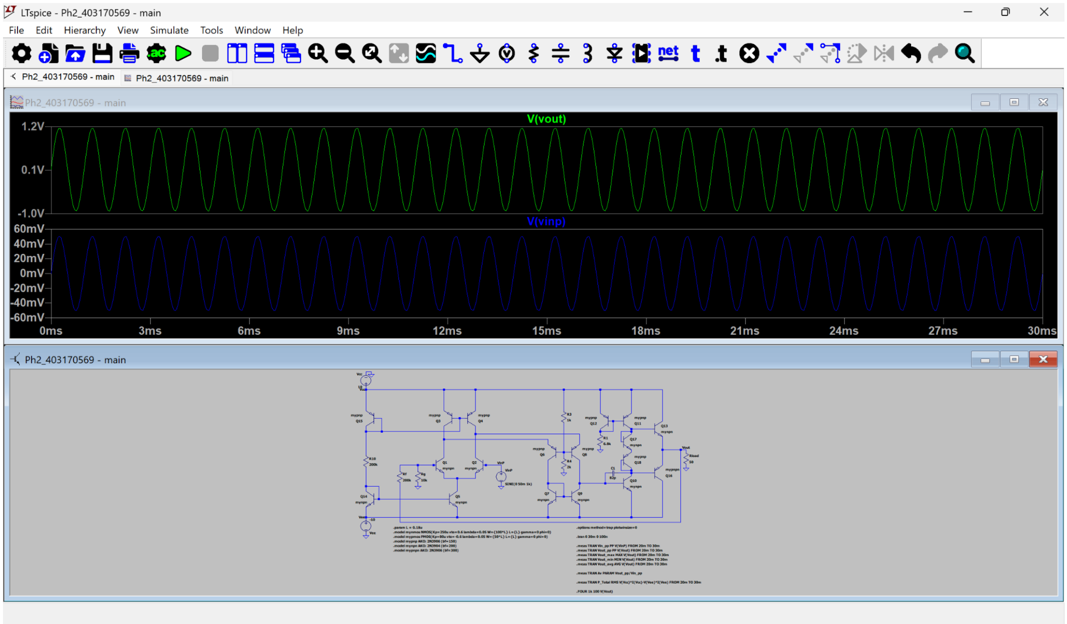

Figure 3 – Input and output waveforms under the standard condition: $V_{in} = 50mV_{peak}$ and $f = 1kHz$.

```
Vin_pp: PP(V(VinP))=0.10000000149 FROM 0.02 TO 0.03
Vout_pp: PP(V(Vout))=2.0877469182 FROM 0.02 TO 0.03
Vout_max: MAX(V(Vout))=1.16226756573 FROM 0.02 TO 0.03
Vout_min: MIN(V(Vout))=-0.925479352474 FROM 0.02 TO 0.03
Vout_avg: AVG(V(Vout))=0.118686570248 FROM 0.02 TO 0.03
Av: Vout_pp/Vin_pp=20.8774688709
P_Total: RMS(V(Vcc)*I(Vcc)-V(Vee)*I(Vee))=0.163758712119 FROM 0.02 TO 0.03
Fourier components of V(Vout)
N-Period=1
DC component:0.118687
```

Figure 4 – Measured gain, output offset, and total power under the standard test condition.

The output is sinusoidal and in phase with the input, with an amplitude approximately 20.88 times larger. The measured gain lies within the permitted range of 18 to 22.

6 Comparison With and Without Feedback

Required Work (Part 5)

Compare the complete circuit with global feedback and without global feedback. The output stage must remain connected in both cases. If the no-feedback circuit saturates, explain why the measured amplitude ratio is not a valid estimate of linear open-loop gain.

Qualitative and Quantitative Comparison

For the no-feedback test, R_f and R_g were removed from the loop and the base of Q_1 was connected directly to ground, while the output stage and the 50Ω load remained connected. The input in both tests was $50mV_{peak}$ at $1kHz$.
With feedback, the output remains sinusoidal with $V_{out,pp} = 2.0877V$ and a gain of 20.877.

Without feedback, the very large open-loop gain drives the output into saturation at approximately $+9V$ and $-9V$ even for this small input, producing an almost square waveform. The ratio of the saturated output amplitude to the input amplitude is not a valid estimate of linear open-loop gain because the output is limited by the supply rails.

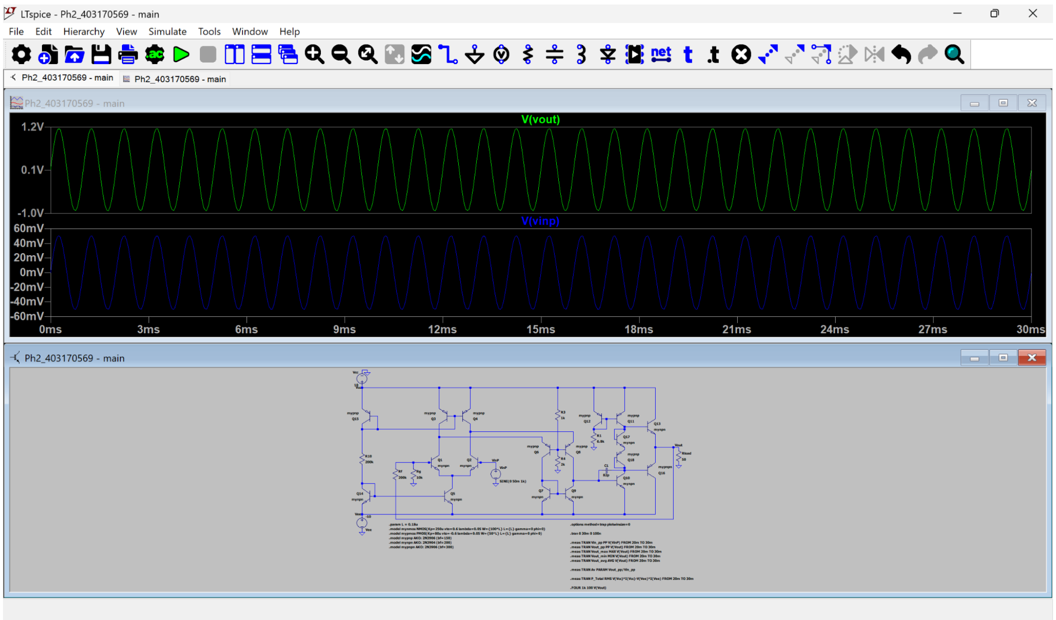

Figure 5 – Amplifier with feedback: linear sinusoidal output for the standard input.

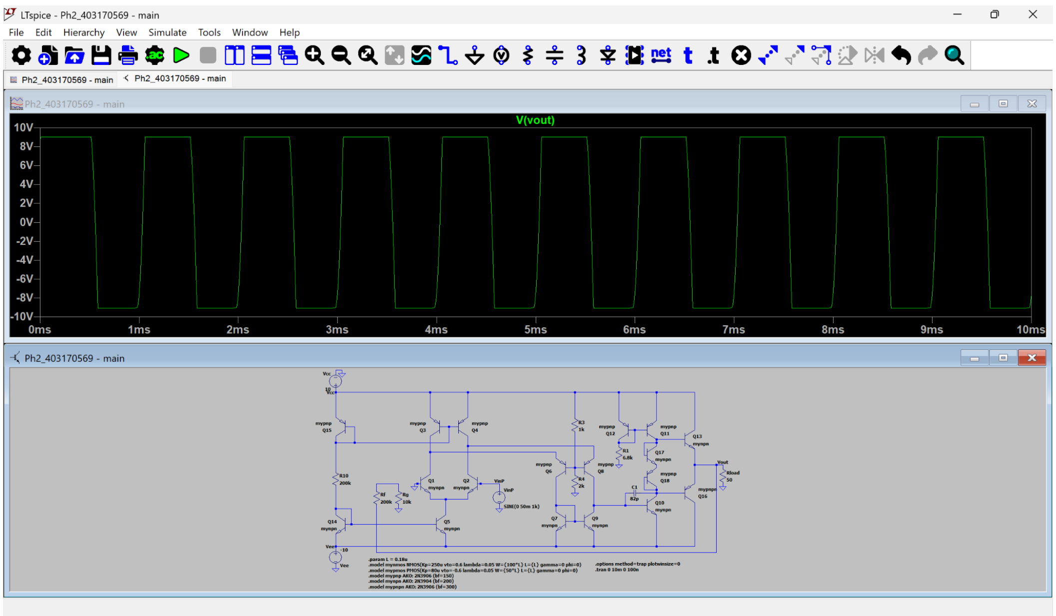

Figure 6 – Amplifier without feedback: severe saturation and clipping for the same input amplitude.

Negative feedback fixes the gain at a predictable value, improves linearity, reduces distortion, and increases the input amplitude required to reach clipping. Without feedback, the amplifier leaves its linear region even for a $50mV$ input.

7 Output Swing, Power, and Efficiency

Required Work (Part 6)

Determine whether the final amplifier can produce a $16V_{pp}$ sinusoidal output across 50Ω without severe clipping or visible distortion. Then measure the load power, total circuit power, and output-stage efficiency.

Theoretical Swing and Efficiency Calculations

With $\pm 10V$ supplies, transistor and driver voltage drops imply that the output peak will be somewhat below each rail. The target $V_p = 8V$ leaves approximately $2V$ of headroom from each supply and is therefore attainable with the emitter-follower topology.
For $V_{out,pp} = 16V$ and $R_L = 50\Omega$,

$$P_{RL,th} = \frac{(V_{pp}/2\sqrt{2})^2}{R_L} = \frac{(16/2\sqrt{2})^2}{50} = 0.64W.$$

Because the closed-loop gain is close to 21, the theoretical required input peak is

$$V_{in,p} \simeq \frac{8}{21} = 0.381V.$$

A value of $385mV$ was used in simulation.
The output-stage efficiency was calculated as

$$\eta_{out} = \frac{P_{RL}}{P_{Q13} + P_{Q16}} \times 100\%.$$

Transient and Power-Measurement Results

For $V_{in,p} = 385mV$ at $1kHz$, the results were

$$\begin{aligned} V_{in,pp} &= 0.7700V, \\ V_{out,pp} &= 16.0501V, \\ V_{out,max} &= 8.1392V, \\ V_{out,min} &= -7.9109V, \\ A_v &= 20.8443. \end{aligned}$$

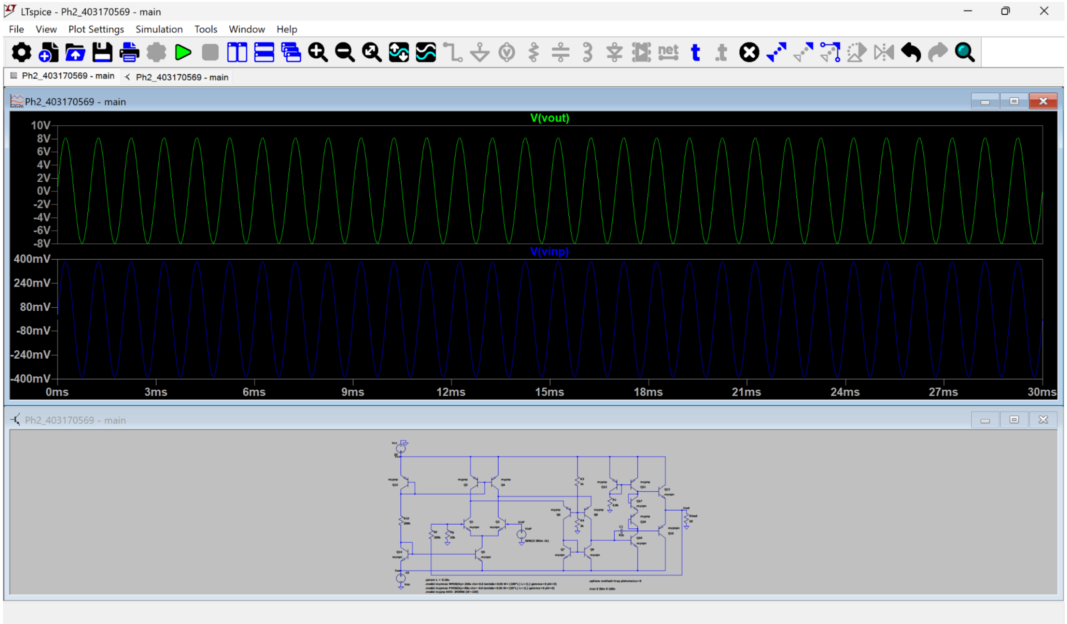

Figure 7 – Final input and output waveforms in the $16V_{pp}$ swing test.

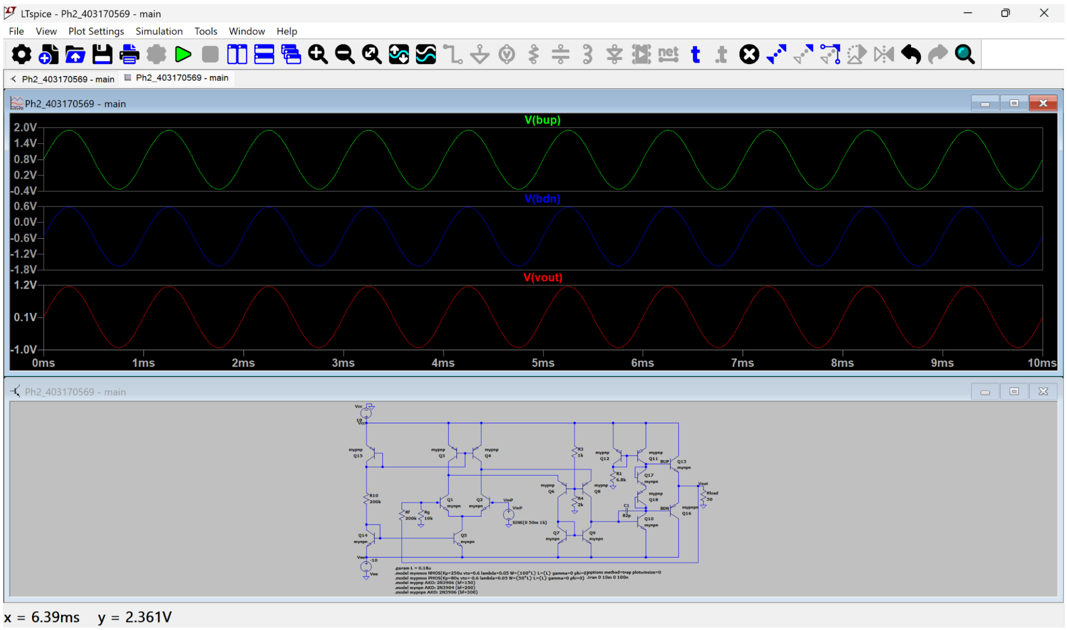

Figure 8 – BUP, BDN, and final output voltages near the required maximum swing.

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SPICE Output Log: C:\Users\sohan\OneDrive\Desktop\university\electronics 2\project\phase2\Ph2_403170569 - main.log
LTspice 24.1.5 for Windows
Circuit: C:\Users\sohan\OneDrive\Desktop\university\electronics 2\project\phase2\Ph2_403170569 - main.net
Start Time: Tue Jul 28 16:55:10 2026
Options: method=trap plotwinsize=0
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = trap
Direct Newton iteration for .op point succeeded.
Total elapsed time: 8.022 seconds.

Files loaded:
C:\Users\sohan\OneDrive\Desktop\university\electronics 2\project\phase2\Ph2_403170569 - main.net
C:\Users\sohan\AppData\Local\LTspice\lib\cmp\standard.bjt

Vin_pp: PP(V(VinP))=0.769999980927 FROM 0.02 TO 0.03
Vout_pp: PP(V(Vout))=16.0500831604 FROM 0.02 TO 0.03
Vout_max: MAX(V(Vout))=8.13917350769 FROM 0.02 TO 0.03
Vout_min: MIN(V(Vout))=-7.91090965271 FROM 0.02 TO 0.03
Vout_avg: AVG(V(Vout))=0.115764961624 FROM 0.02 TO 0.03
Av: Vout_pp/Vin_pp=20.844264361
P_RL_Avg: AVG(V(Vout)*I(Rload))=0.64469097009 FROM 0.02 TO 0.03
P_Q13: AVG(V(Vcc)*Ic(Q13))=0.519963646748 FROM 0.02 TO 0.03
P_Q16: AVG(V(Vee)*Ic(Q16))=0.498021857667 FROM 0.02 TO 0.03
P_out_stage: P_Q13+P_Q16=1.01798550442
Eta_Avg: 100*P_RL_Avg/P_out_stage=63.330073689
```

Figure 9 – Numerical measurements of amplitude, load power, and output-stage efficiency.

Measured values:

- Required input amplitude: $385mV_{peak}$.
- Average load power: $P_{R_L} = 0.644691W$.
- Total circuit power at this operating point: $P_{total} \simeq 1.138W$.
- Upper output transistor power: $P_{Q13} = 0.519964W$.
- Lower output transistor power: $P_{Q16} = 0.498022W$.
- Total output-stage power: $P_{out-stage} = 1.017986W$.
- Efficiency: $\eta_{out} = 63.330\%$, so the 60% requirement is satisfied.

The output remains essentially sinusoidal at this amplitude and no severe clipping is visible. The slight asymmetry between the positive and negative peaks is associated with the DC offset and mismatch between complementary device models. Global feedback shifts the onset of clipping to a substantially larger input amplitude than in the open-loop case.

Main LTspice measurement directives:

```
.meas TRAN Vin_pp  PP V(VinP) FROM 20m TO 30m
.meas TRAN Vout_pp PP V(Vout) FROM 20m TO 30m
.meas TRAN P_RL_Avg AVG V(Vout)*I(Rload) FROM 20m TO 30m
.meas TRAN P_Q13 AVG V(Vcc)*Ic(Q13) FROM 20m TO 30m
.meas TRAN P_Q16 AVG V(Vee)*Ic(Q16) FROM 20m TO 30m
.meas TRAN P_out_stage PARAM P_Q13+P_Q16
.meas TRAN Eta_Avg PARAM 100*P_RL_Avg/P_out_stage
```

8 Total Power Under the Standard Input Condition

Required Work (Part 7)

For a $50mV$-peak, 1 kHz sinusoidal input, measure the total power consumed by the final circuit and compare it with

$$P_{\text{total}} \leq 190mW.$$

Power Measurement Under the Standard Condition

The test was performed with $V_{in} = 50mV_{peak}$, $f = 1kHz$, $\pm 10V$ supplies, and $R_L = 50\Omega$. Total power was calculated from both supply rails over the steady-state interval from $20ms$ to $30ms$.

Parameter	Value
Input amplitude	$50mV_{peak}$
Input frequency	$1kHz$
$V_{out,pp}$	$2.08775V$
Total circuit power	$0.163759W = 163.759mW$
Status	**PASS**

```
Vin_pp: PP(V(VinP))=0.10000000149 FROM 0.02 TO 0.03
Vout_pp: PP(V(Vout))=2.0877469182 FROM 0.02 TO 0.03
Vout_max: MAX(V(Vout))=1.16226756573 FROM 0.02 TO 0.03
Vout_min: MIN(V(Vout))=-0.925479352474 FROM 0.02 TO 0.03
Vout_avg: AVG(V(Vout))=0.118686570248 FROM 0.02 TO 0.03
Av: Vout_pp/Vin_pp=20.8774688709
P_Total: RMS(V(Vcc)*I(Vcc)-V(Vee)*I(Vee))=0.163758712119 FROM 0.02 TO 0.03
Fourier components of V(Vout)
N-Period=1
DC component:0.118687
```

Figure 10 – Total circuit-power measurement for the standard input.

Since $163.759mW < 190mW$, the power-consumption requirement is satisfied with approximately $26.24mW$ of margin.

9 Total Harmonic Distortion Measurement

Required Work (Part 8)

Measure the THD of the final feedback amplifier with the output stage and 50Ω load connected. Use a $50mV$-peak, 1 kHz sinusoidal input. Repeat the test after inserting the specified noise sources in series with the resistors that set the current-source values.

Measurement Method and Interpretation

To remove start-up transients, the transient simulation was run to $30ms$ and Fourier analysis was performed on the steady-state waveform. The maximum time step was $100ns$

and waveform compression was disabled:

```
.options method=trap plotwinsize=0
.tran 0 30m 0 100n
.FOUR 1k 100 V(Vout)
```

In the noisy test, four behavioral sources described by

```
V=(white(2e6*time)/4)
```

were inserted in series with R_{10}, R_3, R_4, and R_1. These resistors set the currents of the input current source, middle stage, and output-bias network.
The added noise randomly modulates the bias currents and creates incoherent spectral components at the output; therefore, the reported total distortion increases. Nevertheless, negative feedback suppresses part of this disturbance.

THD Results

Test Condition	THD	Status
Feedback amplifier, no noise	0.040374%	PASS
Feedback amplifier, with noise sources	0.819073%	PASS

Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [degree]	Normalized Phase [deg]
1	1.000e+03	1.044e+00	1.000e+00	90.37°	0.00°
2	2.000e+03	3.359e-04	3.218e-04	9.73°	-80.63°
3	3.000e+03	1.909e-04	1.829e-04	-174.85°	-265.22°
4	4.000e+03	7.824e-05	7.495e-05	52.35°	-38.01°
5	5.000e+03	1.028e-04	9.847e-05	-167.36°	-257.72°
6	6.000e+03	5.668e-05	5.429e-05	79.38°	-10.98°
7	7.000e+03	5.486e-05	5.256e-05	-165.09°	-255.46°
8	8.000e+03	4.530e-05	4.339e-05	89.76°	-0.60°
9	9.000e+03	2.759e-05	2.643e-05	-162.20°	-252.57°
10	1.000e+04	3.474e-05	3.328e-05	95.00°	4.63°
11	1.100e+04	1.206e-05	1.156e-05	-157.11°	-247.47°
12	1.200e+04	2.548e-05	2.441e-05	98.43°	8.07°
13	1.300e+04	3.636e-06	3.483e-06	-141.53°	-231.89°
14	1.400e+04	1.792e-05	1.716e-05	101.13°	10.77°
15	1.500e+04	1.505e-06	1.442e-06	-29.28°	-119.65°
16	1.600e+04	1.209e-05	1.158e-05	103.51°	13.15°
17	1.700e+04	3.078e-06	2.949e-06	2.98°	-87.38°
18	1.800e+04	7.787e-06	7.460e-06	105.77°	15.41°
19	1.900e+04	3.561e-06	3.411e-06	10.14°	-80.23°
20	2.000e+04	4.749e-06	4.549e-06	108.06°	17.69°
21	2.100e+04	3.359e-06	3.218e-06	13.99°	-76.37°
22	2.200e+04	2.687e-06	2.574e-06	110.54°	20.17°
23	2.300e+04	2.847e-06	2.728e-06	16.83°	-73.54°
24	2.400e+04	1.349e-06	1.292e-06	113.55°	23.18°
25	2.500e+04	2.248e-06	2.154e-06	19.25°	-71.11°
26	2.600e+04	5.288e-07	5.066e-07	118.60°	28.24°
27	2.700e+04	1.678e-06	1.607e-06	21.46°	-68.90°
28	2.800e+04	7.150e-08	6.849e-08	154.14°	63.77°
29	2.900e+04	1.193e-06	1.143e-06	23.59°	-66.77°
30	3.000e+04	1.880e-07	1.801e-07	-74.99°	-165.35°
31	3.100e+04	8.075e-07	7.736e-07	25.70°	-64.67°
32	3.200e+04	2.841e-07	2.721e-07	-67.78°	-158.14°
33	3.300e+04	5.183e-07	4.966e-07	27.80°	-62.56°
34	3.400e+04	2.976e-07	2.851e-07	-64.42°	-154.78°
35	3.500e+04	3.107e-07	2.976e-07	29.96°	-60.41°
36	3.600e+04	2.694e-07	2.581e-07	-61.73°	-152.10°
37	3.700e+04	1.699e-07	1.628e-07	32.45°	-57.91°
38	3.800e+04	2.234e-07	2.140e-07	-59.59°	-149.96°
39	3.900e+04	7.889e-08	7.557e-08	35.31°	-55.05°
40	4.000e+04	1.733e-07	1.660e-07	-57.41°	-147.77°
41	4.100e+04	2.331e-08	2.233e-08	41.57°	-48.80°
42	4.200e+04	1.280e-07	1.226e-07	-55.33°	-145.69°
43	4.300e+04	7.822e-09	7.493e-09	-162.06°	-252.43°
44	4.400e+04	8.926e-08	8.551e-08	-53.40°	-143.76°
45	4.500e+04	2.230e-08	2.136e-08	-146.68°	-237.04°
46	4.600e+04	5.988e-08	5.736e-08	-50.70°	-141.06°
47	4.700e+04	2.751e-08	2.635e-08	-143.01°	-233.37°
48	4.800e+04	3.736e-08	3.579e-08	-49.84°	-140.20°
49	4.900e+04	2.577e-08	2.469e-08	-139.67°	-230.03°
50	5.000e+04	2.157e-08	2.066e-08	-47.49°	-137.86°
51	5.100e+04	2.261e-08	2.166e-08	-137.77°	-228.14°
52	5.200e+04	1.129e-08	1.081e-08	-42.63°	-133.00°
53	5.300e+04	1.872e-08	1.793e-08	-137.06°	-227.43°
54	5.400e+04	3.980e-09	3.813e-09	-45.73°	-136.10°
55	5.500e+04	1.409e-08	1.350e-08	-132.53°	-222.89°
56	5.600e+04	8.081e-10	7.742e-10	-9.52°	-99.89°
57	5.700e+04	1.014e-08	9.717e-09	-133.44°	-223.80°
58	5.800e+04	1.251e-09	1.198e-09	152.78°	62.41°
59	5.900e+04	6.866e-09	6.577e-09	-129.48°	-219.85°
60	6.000e+04	2.504e-09	2.398e-09	141.17°	50.80°
61	6.100e+04	4.393e-09	4.209e-09	-127.36°	-217.73°
62	6.200e+04	2.464e-09	2.361e-09	131.00°	40.64°
63	6.300e+04	2.886e-09	2.765e-09	-126.07°	-216.44°
64	6.400e+04	2.250e-09	2.155e-09	150.50°	60.13°
65	6.500e+04	1.402e-09	1.343e-09	-119.78°	-210.15°
66	6.600e+04	2.359e-09	2.260e-09	145.13°	54.76°
67	6.700e+04	9.255e-10	8.866e-10	-93.62°	-183.99°
68	6.800e+04	1.382e-09	1.324e-09	148.22°	57.85°
69	6.900e+04	4.549e-10	4.358e-10	144.42°	54.05°
70	7.000e+04	1.087e-09	1.042e-09	158.38°	68.01°
71	7.100e+04	2.647e-10	2.536e-10	12.10°	-78.26°
72	7.200e+04	1.187e-09	1.137e-09	145.67°	55.30°
73	7.300e+04	2.232e-10	2.138e-10	-108.33°	-198.70°
74	7.400e+04	3.578e-10	3.428e-10	-122.42°	-212.78°
75	7.500e+04	4.273e-10	4.094e-10	50.72°	-39.64°
76	7.600e+04	1.056e-09	1.011e-09	116.68°	26.32°
77	7.700e+04	5.008e-10	4.798e-10	92.73°	2.36°
78	7.800e+04	3.729e-10	3.573e-10	-14.44°	-104.80°
79	7.900e+04	1.374e-10	1.316e-10	92.29°	1.92°
80	8.000e+04	5.411e-10	5.183e-10	172.17°	81.80°
81	8.100e+04	2.190e-10	2.098e-10	81.93°	-8.43°
82	8.200e+04	2.694e-10	2.581e-10	-40.73°	-131.09°
83	8.300e+04	4.974e-10	4.765e-10	64.97°	-25.40°
84	8.400e+04	5.897e-10	5.649e-10	172.15°	81.78°
85	8.500e+04	4.702e-10	4.504e-10	-94.18°	-184.54°
86	8.600e+04	1.579e-10	1.513e-10	45.19°	-45.17°
87	8.700e+04	3.326e-10	3.186e-10	73.07°	-17.29°
88	8.800e+04	1.565e-10	1.499e-10	-118.04°	-208.41°
89	8.900e+04	3.080e-10	2.950e-10	-79.90°	-170.26°
90	9.000e+04	2.521e-10	2.415e-10	2.44°	-87.93°
91	9.100e+04	2.392e-10	2.292e-10	-170.44°	-260.80°
92	9.200e+04	3.475e-10	3.329e-10	-11.64°	-102.01°
93	9.300e+04	2.739e-10	2.624e-10	44.74°	-45.62°
94	9.400e+04	2.097e-10	2.009e-10	163.98°	73.61°
95	9.500e+04	1.443e-10	1.382e-10	36.00°	-54.36°
96	9.600e+04	3.908e-10	3.743e-10	93.67°	3.30°
97	9.700e+04	6.152e-10	5.893e-10	150.48°	60.12°
98	9.800e+04	3.776e-10	3.617e-10	-53.05°	-143.41°
99	9.900e+04	4.528e-10	4.338e-10	-63.03°	-153.39°
100	1.000e+05	1.724e-10	1.651e-10	100.59°	10.22°

Partial Harmonic Distortion: 0.040374%
Total Harmonic Distortion: 0.040374%

Figure 11 – Fourier table and THD of the final circuit without the added noise sources.

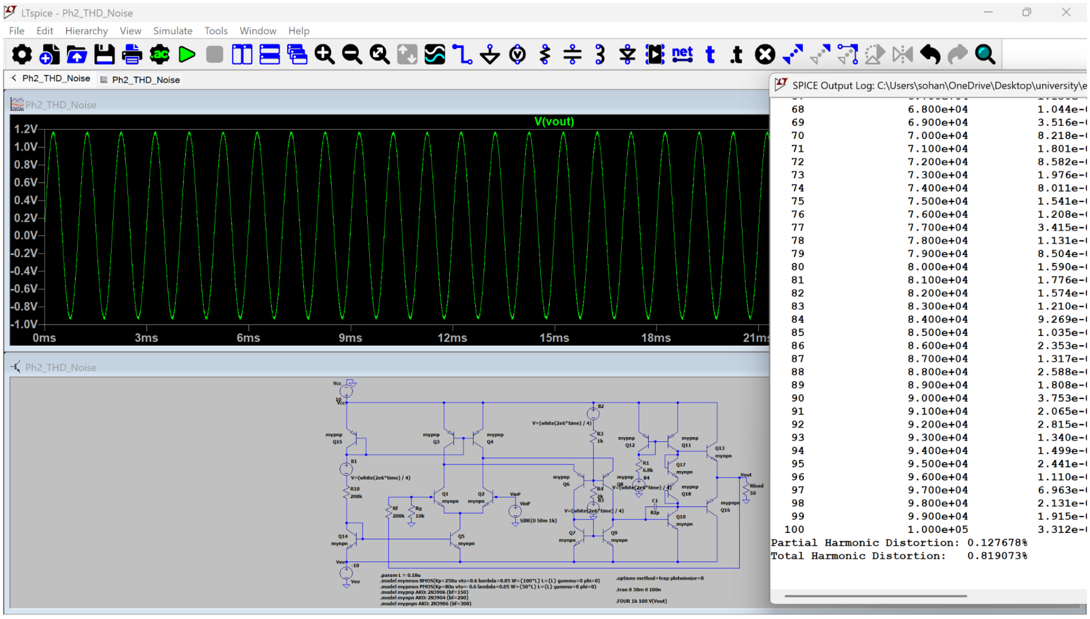


Figure 12 – Noisy circuit, output waveform, and measured $THD = 0.819073\%$.

Without added noise, the second and higher harmonics are extremely small and the THD is approximately 0.0404%. With the noise sources, the total value increases to 0.8191%, which still remains below the 1% limit. In the noisy test, partial harmonic distortion was 0.127678%, while the reported total including noise components was 0.819073%.

10 PSRR Measurement With Sawtooth Supply Ripple

Required Work (Part 9)

Set the signal input to zero and evaluate how supply ripple appears at the output. Use sawtooth sources with $10mV$ ripple for both V_{CC} and V_{EE}, and calculate

$$PSRR = 20\log\left(\frac{V_{\text{ripple}}}{\Delta V_{out}}\right).$$

Test Method and PSRR Calculation

For the PSRR test, the input was set to zero and the DC supplies were replaced by

```
PWL(0 12 5m 12.01 10m 12 15m 12.01 20m 12)
PWL(0 -12 5m -12.01 10m -12 15m -12.01 20m -12)
```

The ripple was defined peak-to-peak, so for each rail

$$V_{ripple,pp} = 10.00002mV.$$

The output peak-to-peak variation from $10ms$ to $20ms$ was

$$\Delta V_{out,pp} = 5.352497 \times 10^{-5}V = 53.525\mu V.$$

Therefore,

$$PSRR = 20\log_{10}\left(\frac{10.00002mV}{53.525\mu V}\right) = 45.429dB.$$

In the LTspice parameter expression, division by $\log(10)$ was used to convert the natural logarithm to base ten.

PSRR Results

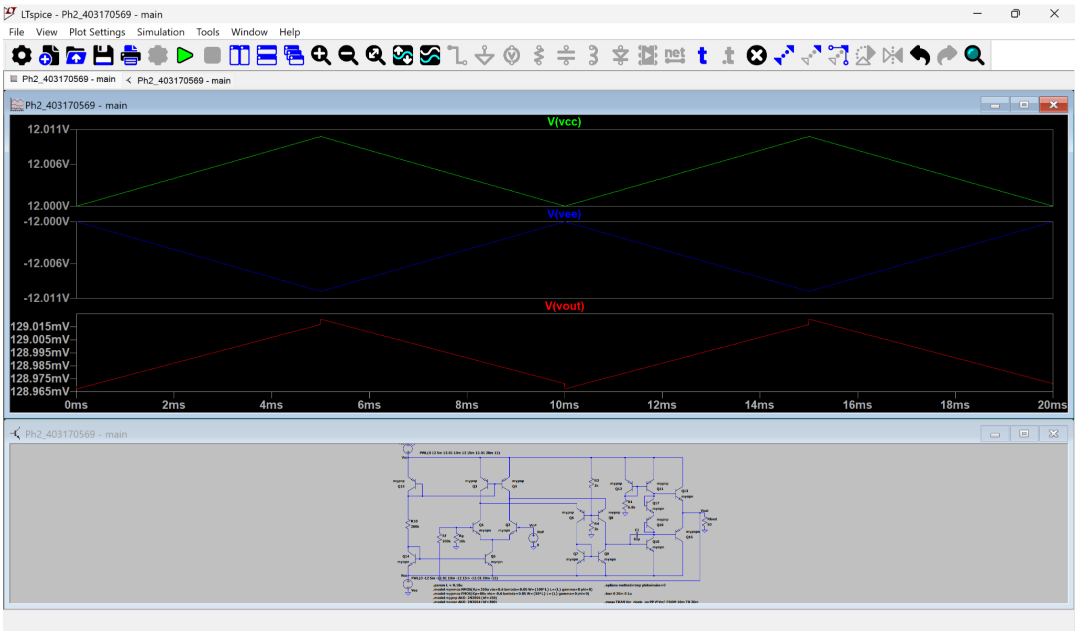
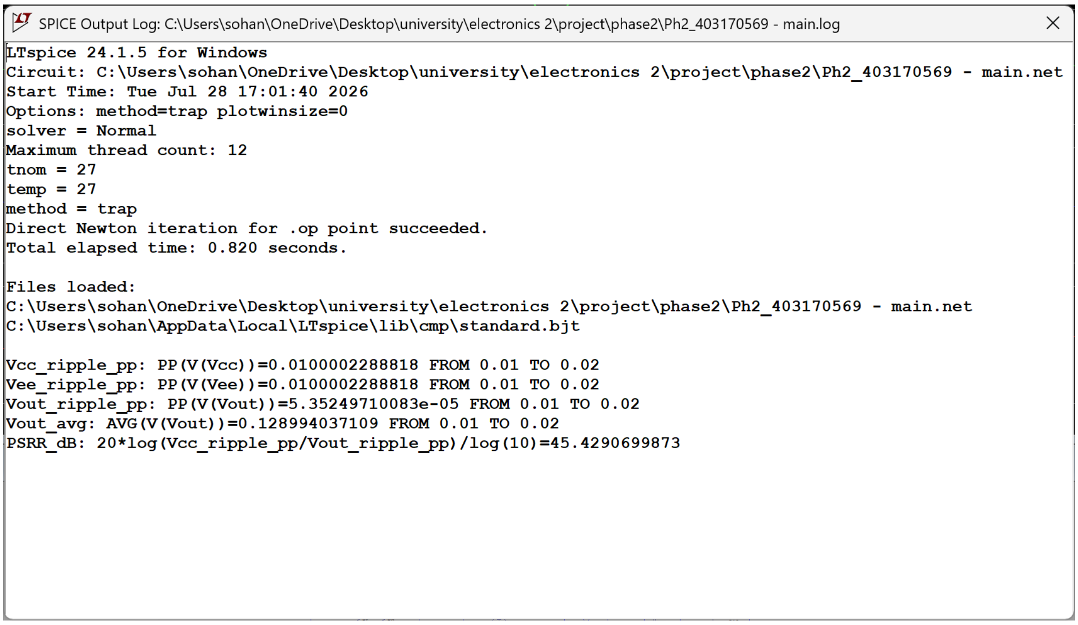


Figure 13 – Sawtooth V_{CC} and V_{EE} waveforms and the much smaller output ripple.



SPICE Output Log: C:\Users\sohan\OneDrive\Desktop\university\electronics 2\project\phase2\Ph2_403170569 - main.log

```
LTspice 24.1.5 for Windows
Circuit: C:\Users\sohan\OneDrive\Desktop\university\electronics 2\project\phase2\Ph2_403170569 - main.net
Start Time: Tue Jul 28 17:01:40 2026
Options: method=trap plotwinsize=0
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = trap
Direct Newton iteration for .op point succeeded.
Total elapsed time: 0.820 seconds.

Files loaded:
C:\Users\sohan\OneDrive\Desktop\university\electronics 2\project\phase2\Ph2_403170569 - main.net
C:\Users\sohan\AppData\Local\LTspice\lib\cmp\standard.bjt

Vcc_ripple_pp: PP(V(Vcc))=0.0100002288818 FROM 0.01 TO 0.02
Vee_ripple_pp: PP(V(Vee))=0.0100002288818 FROM 0.01 TO 0.02
Vout_ripple_pp: PP(V(Vout))=5.35249710083e-05 FROM 0.01 TO 0.02
Vout_avg: AVG(V(Vout))=0.128994037109 FROM 0.01 TO 0.02
PSRR_dB: 20*log(Vcc_ripple_pp/Vout_ripple_pp)/log(10)=45.4290699873
```

Figure 14 – Measured ripple values and calculated PSRR.

Parameter	Value
$V_{ripple,pp}$ on each rail	$10.00002mV$
$\Delta V_{out,pp}$	$53.525\mu V$
$V_{out,avg}$	$128.994mV$
PSRR	$45.429dB$

The output ripple is approximately 187 times smaller than the applied ripple. Thus, at the selected ripple frequency the amplifier strongly suppresses simultaneous changes on the two supply rails. The DC output offset was excluded from the PSRR calculation; only the output peak-to-peak variation was used.

11 Differential Input-Resistance Measurement

Required Work (Part 10)

Measure the differential input resistance of the final circuit and compare it with

$$R_{in,\text{diff}} > 1M\Omega.$$

Explain the measurement procedure.

Analytical Estimate and Measurement Method

In the small-signal model, each input transistor has an approximate base resistance

$$r_\pi = \frac{\beta V_T}{I_C}.$$

For $\beta \simeq 200$ and a collector current of roughly $47\mu A$, the intrinsic value is approximately $110k\Omega$. Global negative feedback keeps the differential input voltage very small and raises the effective closed-loop input resistance by the loop-gain factor, so a value of several tens of megaohms is expected.
For the precise result, LTspice transfer-function analysis was used. LTspice calculates the small-signal input-source current and reports

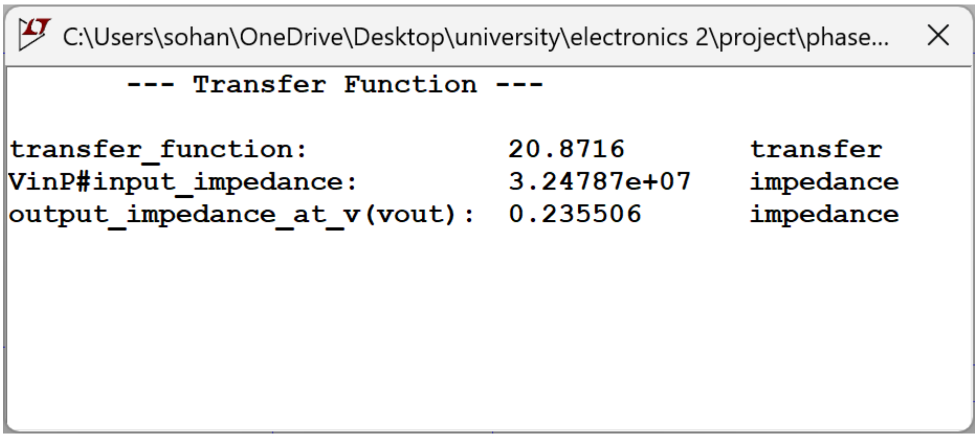
$$R_{in} = \frac{V_{test}}{I_{test}}.$$

Input-Resistance Simulation Result

The final circuit was simulated with $R_L = 50\Omega$ and zero DC input using

```
.tf V(Vout) VinP
```

The $VinP$ source applies the small-signal excitation to the non-inverting input of the differential pair, while the other input is established by the feedback network.



C:\Users\sohan\OneDrive\Desktop\university\electronics 2\project\phase...

```
      --- Transfer Function ---

transfer_function:               20.8716        transfer
VinP#input_impedance:            3.24787e+07    impedance
output_impedance_at_v(vout):     0.235506       impedance
```

Figure 15 – Transfer-function result used to determine the final amplifier input resistance.

- Small-signal gain: 20.8716.
- Effective input resistance: $R_{in,\text{diff}} = 3.24787 \times 10^7 \Omega = 32.4787 M\Omega$.
- Status: **PASS**, because $32.4787 M\Omega > 1 M\Omega$.

12 Output-Resistance Measurement

Required Work (Part 11)

Remove R_L, measure the final circuit output resistance, and compare it with

$$R_{out} < 50\Omega.$$

Explain how the value is extracted from simulation.

Analytical Interpretation

The open-loop output resistance of a complementary emitter follower is on the order of the output-device $1/g_m$ values and driver resistances. Global feedback reduces this resistance approximately by the loop factor:

$$R_{out,CL} \simeq \frac{R_{out,OL}}{1+T}.$$

Thus, the closed-loop value is expected to be much less than 50Ω. The load must be removed so that the external 50Ω resistance is not placed in parallel with the amplifier's intrinsic output resistance.

Output-Resistance Simulation Result

The load was removed completely, the input was set to zero, and the supplies were fixed at $\pm 10V$. The same transfer-function analysis was then run:

```
.tf V(Vout) VinP
```

LTspice applies an internal test source at the output node and reports the voltage-to-current ratio.

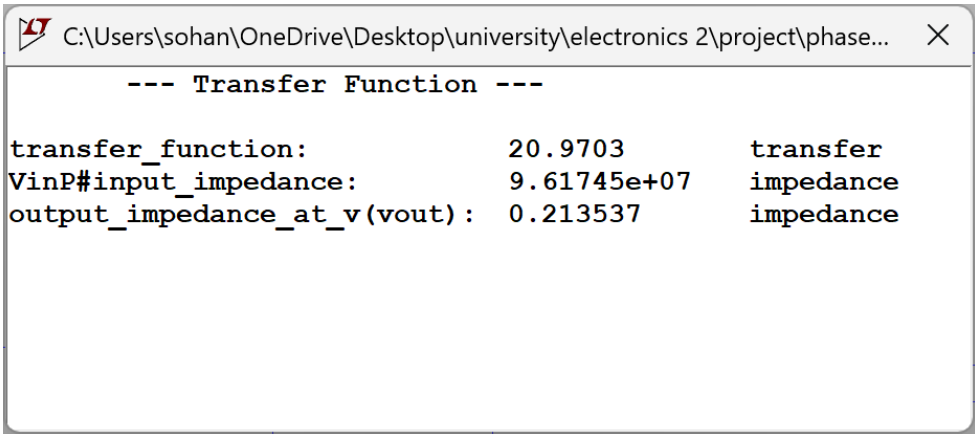


Figure 16 – Output-resistance result after removing R_L.

- $R_{out} = 0.213537\Omega$.
- Unloaded small-signal gain in this analysis: 20.9703.
- Status: **PASS**, because $0.213537\Omega \ll 50\Omega$.

13 Real Audio-Signal Amplification Test

Required Work (Part 12)

Apply a real audio file to the amplifier input and examine the amplified output. A simple sinusoid is not acceptable in place of a real audio signal. Include time-domain input and output waveforms and demonstrate that the output is not severely clipped.

Audio-File Preparation

The original signal was a personal recording named `Soha_audio.m4a`, encoded in AAC at $48kHz$ and two channels. For LTspice, it was converted to

- simulation input filename: `audio.wav`;
- mono, 16-bit PCM;
- sample rate: $44100Hz$;
- duration: $2.665s$;
- normalized amplitude of approximately $40mV_{peak}$ to avoid output clipping.

The input source and output-file directive were

```
wavefile="audio.wav" chan=0
.tran 0 2.665 0 5u
.wave "output.wav" 16 44100 V(Vout)
```

Audio-Test Results

File specifications:

- Input file: `audio.wav`.
- Output file: `output.wav`.
- Both files: $44.1kHz$, mono, 16-bit PCM.

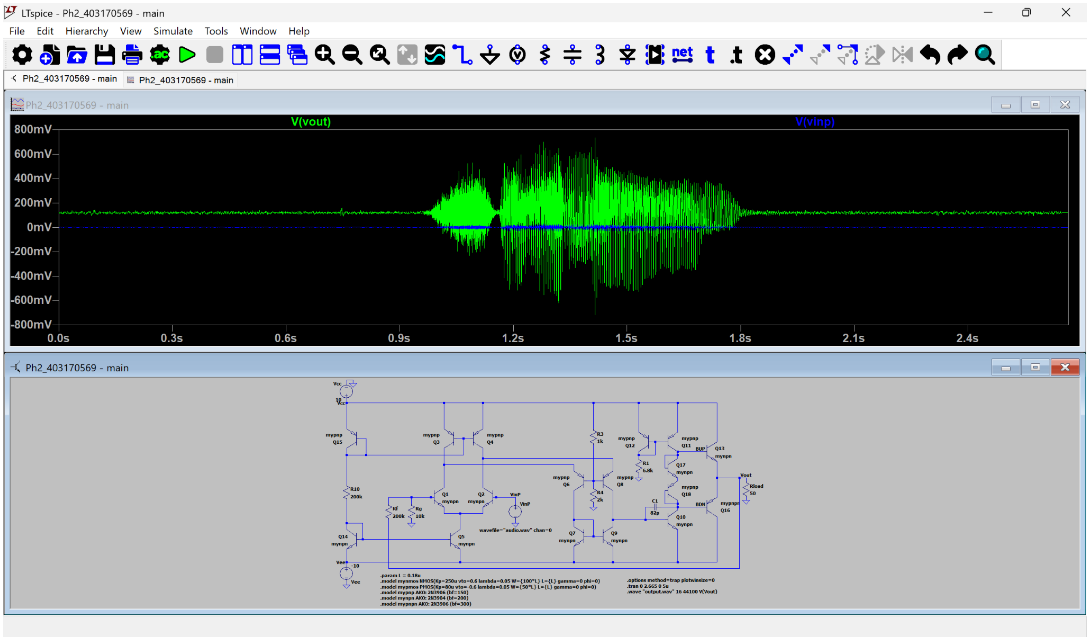


Figure 17 – Time-domain audio input and amplified output in LTspice.

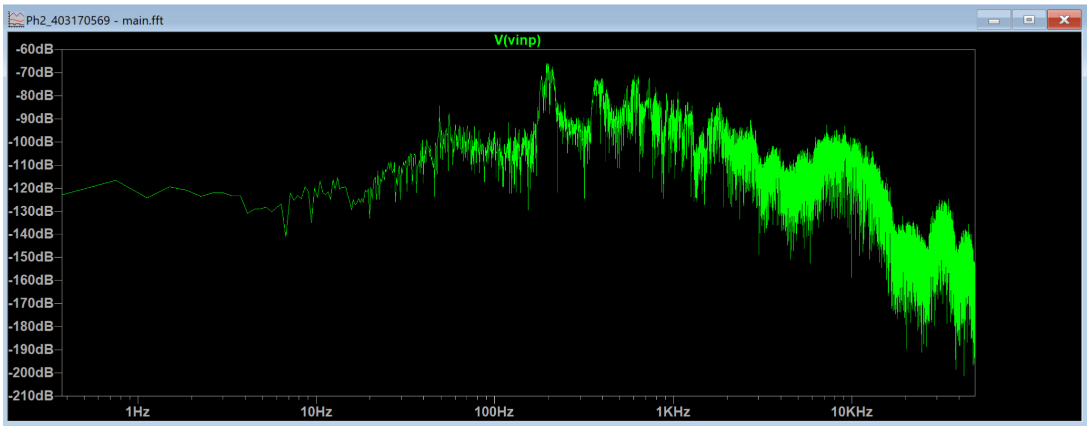


Figure 18 – Frequency spectrum of the input audio file.

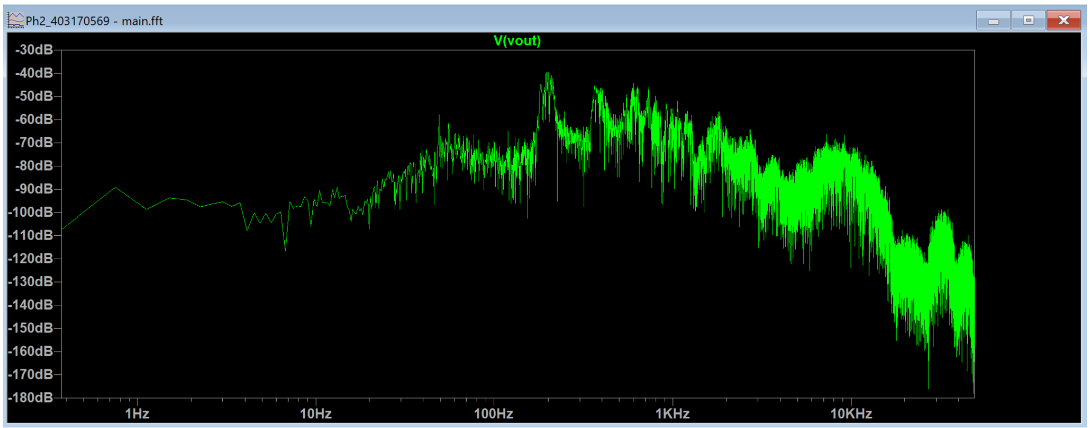


Figure 19 – Frequency spectrum of the amplified output file.

Numerical analysis of the WAV files showed an output range from $-0.716V$ to $+0.734V$, far from either the digital full-scale limit or the circuit supply rails; therefore, severe clipping is absent. The output mean is $0.119V$, and a linear fit between the two files gives a gain of 20.838, or $26.38dB$. The input-output correlation coefficient is approximately 0.9999, and the two spectra retain the same overall shape. Their level difference is approximately equal to the circuit gain, indicating that the principal audio-frequency components are preserved.

Files submitted for this part: `audio.wav` and `output.wav`.

14 Circuit Cost and Component Constraints

Required Work (Part 13)

Determine the final circuit cost using the prescribed cost calculation. If the cost exceeds 200, calculate the penalty from

$$\text{Penalty} = 2\% \times \left\lceil \frac{\text{Cost} - 200}{5} \right\rceil .$$

Component Count and Cost

The load resistor R_L is excluded from the cost calculation as specified. The circuit contains 18 BJTs, six internal resistors, and one capacitor.

Component Type	**Count**	**Unit Cost**	**Total Cost**
MOSFET	0	3	0
BJT	18	5	90
Resistor	6	10	60
Capacitor	1	40	40
Total Cost			190

- Total cost: 190.

- Exceeds the 200 limit: no.
- Penalty: 0%.

The circuit therefore satisfies the project cost limit with a margin of 10 cost units.

15 Optional Bonus Conditions

Required Work (Part 14)

If the optional bonus conditions were evaluated, report the corresponding results here. All three of the following requirements must be met simultaneously to receive bonus credit:

- $17V_{pp}$ output swing without visible clipping;
- efficiency greater than 70% at the same operating point;
- output DC voltage less than $100mV$.

Bonus-Condition Evaluation

Parameter	Required Condition	Measured Value
Output swing	$17V_{pp}$	
Efficiency	$> 70\%$	
$\lvert V_{out,DC}\rvert$	$< 100mV$	

- Corresponding input amplitude: ____________________
- Output waveform under this condition: ____________________
- Final bonus-section result: ____________________

16 Submitted Files

Required Work (Part 15)

Complete the list of files submitted with the project.

Submission Checklist

The final files are placed in the project folder as follows:

- ☐ `Ph2_403170569.asc`: complete final LTspice schematic with the power stage, load, and feedback.
- ☐ `audio.wav`: normalized mono audio input file.
- ☐ `output.wav`: audio output generated by the amplifier simulation.
- ☐ `Elec2_Report_Ph2_403170569_English.pdf`: final English report.

17 Final Conclusion

Required Work (Part 16)

Briefly discuss the main Phase-II design challenge, the most difficult part among the output stage, feedback, THD, PSRR, and audio tests, any unmet specifications, and possible improvements if additional time or cost were available.

Conclusion

The central Phase-II challenge was balancing output swing, efficiency, loop stability, and distortion. Simply adding a push-pull stage to the voltage amplifier was not sufficient: incorrect biasing could produce crossover distortion, while the large loop gain could make the amplifier susceptible to oscillation or undesirable transient behavior. Adjustment of the Q_{17}–Q_{18} bias network and selection of the $C_1 = 82pF$ Miller capacitor were decisive in obtaining a stable response and low THD.

The most difficult task was simultaneous optimization of noisy THD and stability. Increasing the compensation capacitor reduced high-frequency content and improved the noisy THD result, but it was necessary to verify that gain, swing, efficiency, bandwidth, and slew rate remained acceptable. The PSRR test also required a careful peak-to-peak ripple definition and correct base-10 logarithm calculation.

The final circuit satisfies all mandatory specifications: gain 20.877, swing $16.05V_{pp}$, output-stage efficiency 63.33%, standard-condition power $163.76mW$, normal THD 0.0404%, noisy THD 0.819%, input resistance $32.48M\Omega$, and output resistance 0.214Ω.

With additional time and component budget, complementary power transistors with better matching, small emitter resistors for thermal stabilization, a more accurate current source, and an output-offset adjustment circuit could improve symmetry and efficiency. A full AC phase-margin analysis and Monte Carlo simulations for β and temperature variations would further strengthen confidence in the design.